VENN DIAGRAMS

Regions, Overlaps and Choosing the Right Diagram

OAVS TGT | Reasoning (Additional topic from the “etc.” in the syllabus) | Topic 23

Syllabus line (Reasoning, 30 Marks): The verbal and non-verbal topics listed in the syllabus end with “etc.”. This topic is not named in that line, but it is asked in the papers: a numerical Venn-diagram question appears in Social Studies 2023 (Q12).

Built from: OAVS TGT English 2019, English 2022, Mathematics 2024, Science 2024 and Social Studies 2023 (17 May 2023, Shift 2). One Venn-diagram question appears (Social Studies 2023 Q12). It is solved with its original figure. The practice questions were generated or checked by program so that each has exactly one correct option.

What is inside

Ch.	Title	What you get
1	PYQ Analysis	The one PYQ and what it asks
2	Foundation	Region reading, formulas, standard diagrams
3	Solved PYQs	Social Studies 2023 Q12 with its original figure
4	Shortcuts, Traps and Scope	Speed methods and mistakes
5	Practice Set	6 original questions with a verified key
6	Quick Revision	One-page cheat sheet

CHAPTER 1

PYQ Analysis

One question, in Social Studies 2023

Paper	Q.	Question	Figure	Ans
Social Studies 2023	12	How many total doctors are tax-payers also?	Rectangle (Doctors), circle (Females), pentagon (Tax-payers) with numbers	2 (110)
English 2019 / English 2022 / Mathematics 2024 / Science 2024	–	none		

PYQ INSIGHT: The PYQ is a **numerical Venn diagram**: three shapes overlap and each region carries a number. The question asks for the **total** in two sets, so you must add **every region that lies in both**, including the region shared by all three.

Pattern	PYQs	Priority
Numerical Venn: total in two or three sets	1	★★★ Must
Choose the diagram for given classes (Cats, Dogs, Animals)	0	★★★ Expected
Only-one-set and exactly-two-sets counts	0	★★ Expected

CHAPTER 2

Foundation

Regions, formulas and the standard diagrams

2.1 Reading a numerical Venn diagram

Question wording	Add these regions
“in A and B”	A∩B only + A∩B∩C
“in A only”	The region of A that is outside B and C
“in exactly two sets”	A∩B only + B∩C only + A∩C only
“in A or B or both”	Every region of A and B
“in A but not in B”	A only + A∩C only
“in all three”	The centre region

NOTE: A number printed in a region belongs **only to that region**. “Total doctors who are tax-payers” means everything inside both the rectangle and the pentagon, including the area shared with the circle.

2.2 Formulas

Formula	Meaning
$n(A \cup B) = n(A) + n(B) - n(A \cap B)$	Two sets
$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$	Three sets

2.3 Choosing the diagram for given classes

Relation of the classes	Diagram	Example
One class is part of another which is part of a third	Circles inside circles	Mango, Fruits, Food
Two classes inside a third, with nothing in common	Two separate circles inside a bigger circle	Cats, Dogs, Animals
Two classes inside a third, some members in common	Two intersecting circles inside a bigger circle	Doctors, Males, Humans
Three classes with no relation	Three separate circles	Pen, Book, Table
Three classes with some members in common	Three intersecting circles	Students, Girls, Athletes

SHORTCUT: For numbers: list the seven regions in a table and tick the ones the question asks for. For diagrams: ask “is one class part of the other?” then “do they share members?”.

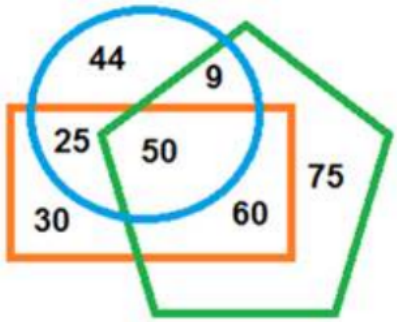
CHAPTER 3

Solved PYQs

Social Studies 2023 Q12 with its original figure

Social Studies 2023 Q12

The rectangle represents ‘Doctors’, the circle represents ‘Females’ and the pentagon represents ‘Tax-payers’. The numbers in different sections indicate the number of persons. How many total doctors are tax-payers also? (1) 60 (2) 110 (3) 50 (4) 75



1. Identify the regions: 44 = females only; 9 = females and tax-payers (not doctors); 25 = females and doctors (not tax-payers); 50 = all three; 30 = doctors only; 60 = doctors and tax-payers (not females); 75 = tax-payers only.
2. Doctors who are also tax-payers lie inside both the rectangle and the pentagon: the region of all three (50) and the region of doctors and tax-payers who are not females (60).
3. Total = 50 + 60 = 110.
4. Option 1 (60) leaves out the region of all three; option 3 (50) leaves out the 60; option 4 (75) is the tax-payers-only region, which is not inside the rectangle.
Answer: Option 2 (110)

CHAPTER 4

Shortcuts, Traps and Scope

Speed methods and mistakes

- **List all regions** before reading the question; most mistakes come from missing the centre region.
- **“Total” means every region inside both sets**, including the centre.
- **Exactly two sets:** do not add the centre.
- **Diagrams:** ask whether one class is inside another (circles inside circles).
- **Time:** 45 seconds per question.

Trap	How to avoid it
Leaving out the centre region	The centre belongs to all three sets.
Adding the centre for “exactly two”	It is in all three, so it is not “exactly two”.
Reading the number outside the circles as a set member	A number outside every shape counts people in none of the sets.
Choosing intersecting circles when one class is inside another	Mango is wholly inside Fruits.

Scope

Related PYQs / topics	Where
Syllogism with Venn diagrams (some, all, no)	Syllogism (earlier topic)
Counting figures	Counting Figures (Topic 21)

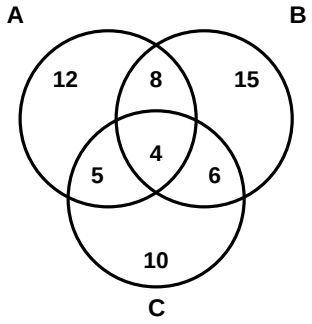
CHAPTER 5

Practice Set

6 original questions; answers on the next page

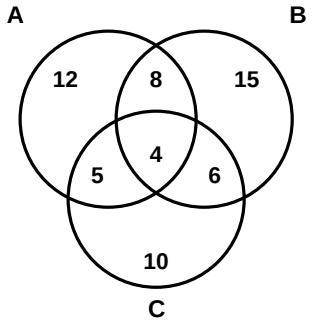
A. Numerical Venn diagram

Q1. In the diagram the numbers show the persons in each region of A, B and C. How many persons are in both A and B (including those also in C)?



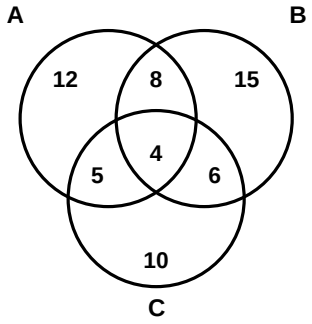
A. 8 B. 12 C. 14 D. 17

Q2. How many persons are in exactly two of the three sets?



A. 23 B. 12 C. 14 D. 19

Q3. How many persons are in set B (anywhere inside B)?



A. 33 B. 15 C. 29 D. 45

B. Choosing the diagram

Q4. Which diagram best represents Cats, Dogs, Animals?



Q5. Which diagram best represents Mango, Fruits, Food?



A B C D

Q6. Which diagram best represents Doctors, Males, Humans?



A B C D

Answer key

Q	Answer	Reason
1	B	Persons in both A and B = region A∩B only (8) + the centre (4) = 12.
2	D	Exactly two = A∩B only (8) + B∩C only (6) + A∩C only (5) = 19. The centre (4) is in all three, so it is not counted.
3	A	B = B only (15) + A∩B only (8) + B∩C only (6) + centre (4) = 33.
4	C	Cats and Dogs are different classes with no members in common, and both are inside Animals: two separate circles inside a bigger circle.
5	A	All mangoes are fruits and all fruits are food: circles inside circles.
6	D	Doctors and Males are both inside Humans, and some doctors are male: two intersecting circles inside a bigger circle.

CHAPTER 6

Quick Revision

One page for the last day

Item	Rule
Total in two sets	Add every region inside both, including the centre
Exactly two	Do not add the centre
Three-set union	Sum − pairs + triple
Part-of relation	Circle inside circle
Overlap without containment	Intersecting circles

PYQ RULE: Social Studies 2023 Q12: doctors who are tax-payers = region of all three (50) + doctors and tax-payers only (60) = 110.

ROUTINE: Exam-day routine: label the regions; decide which regions the question wants; add them; check the centre. 45 seconds per question.